

Jonas Lim Hong Xiang

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Software and ML engineer with 7 years delivering production AI systems end-to-end across automotive, semiconductor, and SaaS domains. Hands-on experience building GenAI and Agentic AI applications, designing scalable Python microservices, and establishing engineering standards for model evaluation and deployment. Proven track record of owning solutions from design through production adoption while coordinating closely with cross-functional business and technical stakeholders.

PROFESSIONAL EXPERIENCE

Co-Founder & Software Architect

May 2024 - Jun 2026

NockHub, Singapore | nockhub.com

- Architected a fully serverless Python backend on AWS with 40+ data schemas and 30+ production APIs, owning the full solution lifecycle from design to deployment.
- Evaluated and integrated GenAI tools into product workflows; defined internal guidelines for model selection, data handling, and responsible use across automated pipelines.
- Gathered requirements from non-technical stakeholders, structured user acceptance testing cycles, and incorporated feedback to improve solution adoption.
- Defined engineering standards and delivery governance across a team of 3 engineers to maintain consistent quality across product workstreams.
- Grew platform to 4 ranges, 7,000+ users, and 4,000+ bookings per month.

Senior Machine Learning Engineer

Jun 2022 - Jul 2024

Carro, Singapore

- Owned end-to-end delivery of production ML systems across predictive pricing, vehicle damage detection, and engine acoustics for multi-market automotive operations.
- Built scalable Python ML microservices on GPU clusters serving 12,000+ requests per day; designed for reliability, low latency, and maintainability.
- Led model selection, technical assessment, and production transition from Mask R-CNN to YOLOv5 based on quality, latency, and cost requirements; reduced inference time from 500ms to 50ms.
- Established model evaluation frameworks, data quality standards, and a metadata store; improved training data quality by 50%, enabling repeatable and auditable model validation.
- Implemented automated ML pipelines and monitoring practices, reducing operational overhead by 75 engineer-hours per month and improving deployment reliability.
- Partnered with product, operations, and data teams across Singapore, Malaysia, Indonesia, and Taiwan to validate solution behaviour, surface improvement opportunities, and drive adoption.

Data Scientist

Jun 2019 - May 2022

Micron Technology, Singapore

- Built and deployed a predictive yield model within on-premises compliance constraints, contributing to USD 11M in cost savings.
- Built a defect classification model within semiconductor quality controls; improved inspection tool capacity by 10%.
- Delivered end-to-end ML solutions from problem framing to production deployment, collaborating with engineering, quality, and operations teams.

EDUCATION & AWARDS

Master of Computer Science (Data Science & AI)

Aug 2024 - Jun 2026

University of Sydney, Australia

Distinction | Singapore IMDA SG Digital Scholar

- Capstone: Designed and deployed an AI-powered Tour Guide on a Unitree Go2 quadruped robot – built LLM-driven agent workflows for real-time natural language interaction, prompt orchestration, and autonomous task execution.

Bachelor of Engineering (Aerospace)

Aug 2015 - May 2019

Nanyang Technological University, Singapore

First Class Honors, Highest Distinction

TECHNICAL EXPERTISE

GenAI & LLM: Agentic AI, prompt engineering, agent-based workflows, RAG pipelines

Languages: Python, SQL, TypeScript, Dart, Bash

Cloud & Infrastructure: AWS, GCP, Terraform, Firebase

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, RDS

ML/AI: PyTorch, TensorFlow, model evaluation & versioning

Backend & APIs: FastAPI, Docker, production microservices

Data Engineering: PySpark, automated ML pipelines

Robotics & Middleware: ROS 2

HACKATHONS

2025 Global Health Systems Hackathon Hub, "Building High-Value Health Systems: Leveraging AI". Team I Know You Know won 2nd place in Sydney and placed Top 20 globally.

CERTIFICATIONS

AWS Certified Solutions Architect - Professional

Jan 2024

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